Module 1: Differential Equations of First Order and First Degree

Practice Problems in Formulation

1. (Half Life of Uranium) Uranium is an alternate source of fuel for the generation of electricity. It exhibits the following property: It disintegrates at a rate proportional to the amount present at any given instant.
   If m_1 and m_2 g of Uranium are present in the furnace at time t_1 and t_2 respectively, find the half life of Uranium

2. (Antenna Design) A reflector reflects signals in parallel, when these signals are coming from a point source. A receiver receives these parrallel signals and converges to a point when these signals are coming from a long distance. Design a shape of the reflector /receiver for the antenna in a communication system

3. (Escape Velocity) A satellite is launched via a rocket to a particular orbit. Find the initial velocity of the satellite which is fired via a multi-stage rocket in radial direction from the earth's centre and is supposed to escape the earth.

4. (Newton's Law of Cooling) Newton's Law of Cooling states that the rate at which a substance cools in moving air is proportional to the difference between the temperature of the substance and the temperature of the air. Formulate the differential equation and obtain the temperature T of a substance at any time t.

5. ($L-R$ circuit) A resistance R and an inductance L are connected in a series with a voltage supply $E(t)$. Find the current in the circuit.

6. ($R-C$ circuit) A resistance R and a Capacitor C are connected in a series with a voltage supply $E(t)$. Find the current in the circuit.

7. A person deposits 1000 in a bank with 10% compound interest over 5 years. What is the return value after 5 years?

8. A population grows in a country at the rate of r_0 per year. How long does it take for the population to double?

Q1) Solve the following Differential Equations

1. $\left(y^2e^{xy^2}+4x^3\right)dx+\left(2xye^{xy^2}-3y^2\right)dy=0$

2. $\dfrac{y}{x^2}\cos\left(\dfrac{y}{x}\right)dx-\dfrac{1}{x}\cos\left(\dfrac{y}{x}\right)dy+2xdx=0$

3. $(x^4e^x-2mxy^2)\,dx+(2mx^2y)\,dy=0$

4. $\left(1+e^{x/y}\right)dx+e^{x/y}\left(1-\dfrac{x}{y}\right)dy=0$ given $y\left(0\right)=4$
5. $y\left(1+xy\right)dx+x\left(1+xy+x^2y^2\right)dy=0$
6. $\dfrac{ydx-xdy}{\left(x+y\right)^2}=\dfrac{dx}{2\sqrt{1-x^2}}$
7. $(x^4+y^4)dx-xy^3dy=0$
8. $\dfrac{dy}{dx}=\dfrac{y+1}{\left(y+2\right)e^y-x}$
9. $y\left(xy+e^x\right)dx-e^xdy=0$
10. $y-x\dfrac{dy}{dx}=x+y\dfrac{dy}{dx}$
11. $\dfrac{dy}{dx}+\dfrac{x^2y^3+2y}{2x-2x^3y^2}=0$
12. $\left(x^2-3xy+2y^2\right)dx+x(3x-2y)dy=0$
13. $xe^x\left(dx-dy\right)+e^xdx+ye^ydy=0$

Q2) Solve the following Differential Equations

1. $(x^2-1)\sin x\dfrac{dy}{dx}+\left[2x\sin x+\left(x^2-1\right)\cos x\right]y=\left(x^2-1\right)\cos x$
2. $4x^2y\dfrac{dy}{dx}=3x\left(3y^2+2\right)+\left(3y^2+2\right)^3$
3. $4xy\dfrac{dy}{dx}=\left(y^2+3\right)+x^3\left(y^2+3\right)^3$
4. $\dfrac{dr}{d\theta}=r\tan\theta-\dfrac{r^2}{\cos\theta}$
5. $\left(1+y^2\right)dx=\left(\tan^{-1}y-x\right)dy$
6. $\dfrac{dx}{dy}=e^{y-x}\left(e^y-e^x\right)$
7. $\sin 2x\dfrac{dy}{dx}=y+\tan x$
8. $\tan y\dfrac{dy}{dx}+\tan x=\cos y.\cos^3 x$
9. $e^y\left(y+1\right)dy+\left(x^2e^{2x}-ye^y\right)dx=0$
10. $y\dfrac{dx}{dy}=x-yx^2\sin y$
11. $x\dfrac{dy}{dx}+y=y^3x^{n+1}$

Module 2: Linear Differential Equations Of Higher Order With Constant Coefficients

Q.1) Solve the following equations:

1. $(D^2 - 2D - 4)y = 0$
2. $\dfrac{d^4y}{dx^4} + k^4y = 0$
3. $(D^2 + 1)^3(D^2 + D + 1)^2y = 0$
4. $(D - 1)^4(D^2 + 2D + 2)^2y = 0$
5. $(D^4 - 4D^3 + 8D^2 - 8D + 4)y = 0.$
6. $(D^2 + 4D + 4)y = x.e^{-2x}. \cos x$
7. $(D^3 + 2D^2 + D)y = x^2 + x$
8. $(D^4 - 1)y = \cos x. \cosh x$
9. $\operatorname{cosec} x \dfrac{d^4y}{dx^4} + (\operatorname{cosec} x)y = \sin 2x$
10. $(D^2 + 3D + 2)y = e^{e^x}$
11. $(D^2 + 3D + 2)y = x.e^x. \sin x$
12. $(D^2 - 4D + 4)y = 8(x^2 + \sin x + e^{2x})$
13. $(D^3 - D)y = 2e^x + 2x + 1 - 4 \cos x$
14. $(D^2 + 3D + 2)y = \sin(e^x)$
15. $(D^2 - 4)y = x. \sinh x$
16. $(D^2 + 2)y = x. \sin 3x + x^2e^{3x}$
17. $(D^2 + 1)y = 2^x + \sin x. \sin 2x$

Q.2) Solve by using Method of variation of Parameters:

1. $\dfrac{d^2y}{dx^2} - y = \dfrac{2}{1 + e^x}$
2. $\dfrac{d^2y}{dx^2} + y = \sec x. \tan x$
3. $\dfrac{d^2y}{dx^2} + y = \dfrac{1}{1 + \sin x}$
4. $(D^2 - 6D + 9)y = \dfrac{e^{3x}}{x^2}$
5. $\dfrac{d^2y}{dx^2} - 4\dfrac{dy}{dx} + 4y = e^{2x} \sec^2 x$
6. $(D^3 + D)y = cosecx$
7. $(D^2 - 2D + 2)y = e^x \tan x$
8. $(D^2 - 1)y = e^{-x} \sin e^{-x} + \cos e^{-x}$

Module 3: Beta and Gamma Functions, Differentiation Under the Integral Sign (DUIS)

1. Evaluate

   (a) $\int_0^\infty (x^2+4)e^{-2x^2}dx$

   (b) $\int_0^1 \frac{dx}{\sqrt{-\log x}}$

   (c) $\int_0^\infty x^2 7^{-4x^2}dx$

   (d) $\int_0^\infty 3^{-4x^2}dx$

   (e) $\int_0^\infty \frac{x^2}{(1+x^6)^{5/2}}dx$

   (f) $\int_0^\infty \frac{x^3(x^3-1)}{(1+x^3)^5}x^2dx$

   (g) $\int_3^7 \sqrt[4]{(x-3)(7-x)}dx$

   (h) $\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} (cos\theta + sin\theta)^{1/3}d\theta$

2. Show that $\int_0^1 \frac{x^2(4-x^4)}{\sqrt{1-x^2}}dx = 2\beta\left(\frac{3}{2},\frac{1}{2}\right) - \frac{1}{2}\beta\left(\frac{7}{2},\frac{1}{2}\right)$

3. Prove that $\int_0^a \frac{dx}{(a^n-x^n)^{1/n}} = \frac{\pi}{n}cosec\left(\frac{\pi}{n}\right)$.

4. Prove that $\int_0^{\pi/2} \sqrt{sin\theta}d\theta.\int_0^{\pi/2} \frac{d\theta}{\sqrt{sin\theta}} = \pi$

5. Prove that $\int_0^3 \frac{x^{3/2}}{\sqrt{3-x}}dx.\int_0^1 \frac{dx}{\sqrt{1-x^{1/4}}} = \frac{432}{35}\pi$.

6. Prove that $\int_0^{\pi/2} \sqrt{tan\theta}d\theta.\int_0^{\pi/2} \sqrt{cot\theta}d\theta = \frac{\pi^2}{2}$

7. Prove that $\int_0^\infty \frac{e^{-x}-e^{-ax}}{xsecx}dx = \frac{1}{2}\log\left(\frac{a^2+1}{2}\right)$.

8. Prove that $\int_0^\infty e^{-(x^2+\frac{a^2}{x^2})}dx = \frac{\sqrt{\pi}}{2}e^{-2a}$, a > 0

9. Prove that $\int_0^\infty \frac{tan^{-1}ax}{x(1+x^2)}dx = \frac{\pi}{2}\log(1+a)$

10. Prove that $\int_0^\infty \frac{1-\cos ax}{x}e^{-x}dx = \frac{1}{2}\log(a^2+1)$

11. Show that $\int_0^\pi \frac{\log(1+a\cos x)}{\cos x}dx = \pi\sin^{-1}a$, $0 \le a \le 1$

Module 4: Multiple Integration-I

1. Evaluate the following Integrals

(a) $\int_0^1 \int_0^y xye^{-x^2} dxdy$

(b) $\int_0^1 \int_0^1 \frac{dxdy}{\sqrt{1-x^2}\sqrt{1-y^2}}$

(c) $\int_0^{\pi/4} \int_0^{\sqrt{\cos 2\theta}} \frac{r}{(1+r^2)^2} drd\theta$

(d) $\int_0^{\pi/2} \int_0^{a\cos\theta} r\sqrt{a^2-r^2} drd\theta$

2. Sketch the area of integration and evaluate:

(a) $\int_1^2 \int_{-\sqrt{2-y}}^{\sqrt{2-y}} 2x^2y^2 dxdy$

(b) $\int_0^1 \int_0^{y^2} 3y^3 e^{xy} dxdy$

(c) $\int_1^4 \int_0^{\sqrt{x}} \frac{3}{2} e^{y/\sqrt{x}} dydx$

(d) $\int_0^\infty \int_x^\infty e^{-y} dydx$

3. Change the order of integration for the following integrals:

(a) $\int_0^{2a} \int_{\sqrt{2ax-x^2}}^{\sqrt{2ax}} f(x,y) dxdy$

(b) $\int_0^1 \int_{\sqrt{2x-x^2}}^{1+\sqrt{1-x^2}} f(x,y) dxdy$

(c) $\int_0^\pi \int_0^x \frac{\sin y}{\sqrt{(\pi-x)(x-y)}} dxdy$

(d) $\int_0^1 \int_0^{\sqrt{1-y^2}} \frac{\cos^{-1} x}{\sqrt{1-x^2}\sqrt{1-x^2-y^2}} dxdy$

(e) $\int_0^5 \int_{2-x}^{2+x} dydx$

4. Change to polar cordinates and evaluate:

(a) $\int_0^a \int_{2\sqrt{ax}}^{\sqrt{5ax-x^2}} \frac{xdydx}{\sqrt{x^2+y^2}}$

(b) $\int_0^4 \int_{\sqrt{4x-x^2}}^{\sqrt{16-x^2}} \frac{dxdy}{\sqrt{16-x^2-y^2}}$

(c) $\int_0^a \int_{\sqrt{ax-x^2}}^{\sqrt{a^2-x^2}} \frac{dxdy}{\sqrt{a^2-x^2-y^2}}$

(d) $\int_0^a \int_0^{\sqrt{a^2-x^2}} y(\sqrt{x^2+y^2}) dxdy$

5. Evaluate $\iint_R (x^2-y^2)xdxdy$ over the area of the triangle whose vertices are at the points (0,1), (1,1), (1,2). Evaluate $\iint xy\sqrt{1-x-y}dxdy$ over the area of the triangle formed by $x=0, y=0$, and $x+y=1$.

6. Evaluate $\iint r\sqrt{a^2-r^2} drd\theta$ over the upper half of the circle $r = a\cos\theta$.

7. Calculate $\iint r^3 drd\theta$ over the area included between the circles $r = 2\sin\theta$ and $r = 4\sin\theta$.

8. Change to polar cordinates and evaluate $\iint \sqrt{\dfrac{1-x^2-y^2}{1+x^2+y^2}}dxdy$, over the area of the +ve quadrant of the circle $x^2+y^2=1$.

9. Evaluate $\iint \dfrac{(x^2+y^2)^2}{x^2y^2}dxdy$ over the area common to $x^2+y^2=ax$ and $x^2+y^2=by$, $a,b>0$.

10. Evaluate $\iint \dfrac{rdrd\theta}{(r^2+a^2)^2}$ over one loop of the lemniscate $r^2=a^2\cos 2\theta$.

Module 5: Multiple Integration-II

1. Evaluate the following Integrals

   (a) $\displaystyle\int_0^2\int_0^x\int_0^{2x+2y} e^{x+y+z}dxdydz.$

   (b) $\displaystyle\int_0^1\int_0^{1-x}\int_0^{1-x-y}\frac{1}{(1+x+y+z)^3}dxdydz.$

   (c) $\displaystyle\int_0^{\pi/2}\int_0^{a\cos\theta}\int_0^{\sqrt{a^2-r^2}} rdzdrd\theta.$

   (d) $\displaystyle\int_0^\infty\int_0^\infty\int_0^\infty\frac{1}{(1+x^2+y^2+z^2)^2}dxdydz.$

   (e) $\displaystyle\int_0^1\int_0^{\sqrt{1-x^2}}\int_{\sqrt{x^2+y^2}}^1\frac{1}{\sqrt{x^2+y^2+z^2}}dxdydz.$

2. $\iiint \dfrac{1}{(1+x+y+z)^3}dxdydz$ over the volume bounded by the planes $x=0, y=0, z=0$ and $\dfrac{x}{a}+\dfrac{y}{b}+\dfrac{z}{c}=1$.

3. $\iiint (x^2+y^2+z^2)dxdydz$ over the first octant of the sphere $x^2+y^2+z^2=a^2$.

4. $\iiint \sqrt{1-\dfrac{x^2}{a^2}-\dfrac{y^2}{b^2}-\dfrac{z^2}{c^2}}dxdydz$ throughout the volume of the ellipsoid $\dfrac{x^2}{a^2}+\dfrac{y^2}{b^2}+\dfrac{z^2}{c^2}=1$

5. $\iiint x^2yzdxdydz$ over the volume of the tetrahedron bounded by $x=0, y=0, z=0$ and $x+y+z=1$.

6. Find the volume bounded by the cordinate planes and the plane $x+y+z=1/6$.

7. Evaluate $\iiint \dfrac{dxdydz}{(x^2+y^2+z^2)^{1/2}}$ over the volume bounded by the spheres $x^2+y^2+z^2=a^2$ and $x^2+y^2+z^2=a^2$, a,b>0.

8. Find the volume of the tetrahedron bounded by the planes $x=0$, $z=0, x=2y$ and $x+2y+z=2$.

9. Find the volume of the paraboloid $x^2 + y^2 = 4z$ cut off by the plane $z = 4$.

10. Using triple integration find the volume cut off from the sphere $x^2 + y^2 + z^2 = 16$ by the plane $z = 0$ and the cylinder $x^2 + y^2 = 4x$.

Module 6: Numerical solution of ODE of First Order and First Degree and Numerical Integration

1. Find the value of the integral $\int_0^1 \frac{x^2}{1+x^3}dx$, using (i) Trapezoidal rule (ii)Simpson's $1/3^{rd}$ rule. Compare it with the exact value of the integral. Also find the value of $\log 2$.

2. Find the value of the integral $\int_0^6 \frac{dx}{1+x^2}$, using Simpson's $3/8^{th}$ rule. Also find the error.

3. Use Simpson's $1/3^{rd}$ rule to evaluate $\int_1^2 \frac{dx}{x}$ taking 10 equal intervals.

4. In an experiment a quantity G was measured as follows: G(20)=95.9, G(21)=96.85, G(22)=97.77, G(23)=98.68, G(24)=99.56, G(25)=100.41, G(26)=101.24. Compute $\int_{20}^{26} G(x)dx$ by both Simpson's $1/3^{rd}$ rule and Trapezoidal rule.

5. Evaluate $\int_{-1}^1 \frac{dx}{1+x^2}$ using (i) Trapezoidal rule (ii)Simpson's $1/3^{rd}$ rule and (iii) Simpson's $3/8^{th}$ rule. Also find the exact value.

6. Using Euler's Method, find the approximate value of 'y' for the D.E, $\frac{dy}{dx} = \frac{y-x}{\sqrt{xy}}$ when $x = 1.5$ in five steps by taking $h = 0.1, y(1) = 2$

7. Using Euler's Method, find the approximate value of 'y' correct upto 4 decimal places for$x = 0.1$ given $\frac{dy}{dx} = x - y^2$ at $x = 0, y = 1$.Take $h = 0.02$

8. Solve $\frac{dy}{dx} = 2 + \sqrt{xy}$ with initial conditions $y(1.2) = 1.6403$ by using Euler's Modified method. Also find 'y' at $x = 1.4$ and$x = 1.6$. Correct upto 4 decimal places.Take $h = 0.2$

9. Solve $\frac{dy}{dx} = x - y^2$ with initial conditions $y(0) = 1$ by using Euler's Modified method. Also find 'y' at $x = 0.05$ *and* $x = 0.1$. Correct upto 4 decimal places.

10. Apply Runge Kutta Method of fourth order to find an approximate value of 'y' if $\frac{dy}{dx} = xy$, given $y(1) = 1$ for $x = 1.2$.Take $h = 0.1$

11. Apply Runge Kutta Method of fourth order to find an approximate value of 'y' if $\frac{dy}{dx} + y + xy^2 = 0$, given $y(0) = 1$ for $x = 0.2$.Take $h = 0.1$. Correct upto 4 decimal places.

The Principal Formulae

Trigonometry

1. $sin^2\theta + cos^2\theta = 1,\ sec^2\theta = 1 + tan^2\theta,\ cosec^2\theta = 1 + cot^2\theta$

2. $sin0^o = 0,\ cos0^o = 1,\ sin45^o = cos45^o = \frac{1}{\sqrt{2}}, sin60^o = \frac{\sqrt{3}}{2},$
$\cos60^o = \frac{1}{2},\ sin90^o = 1,\ cos90^o = 0$

3. $sin(-\theta) = -sin\theta, \quad sin(180^o - \theta) = sin\theta, \quad cos(-\theta) = cos\theta \quad cos(180 + \theta) = -cos\theta$
$sin(90^o - \theta) = cos\theta, \quad sin(180 + \theta) = -sin\theta, \quad cos(90^o - \theta) = sin\theta \quad cos(180^o - \theta) = -cos\theta$
$sin(90^o + \theta) = cos\theta, \quad cos(90^o + \theta) = -sin\theta$

4. $sin(A \pm B) = sinA.cosB \pm cosA.sinB \qquad cos(A \pm B) = cosA.cosB \mp sinA.sinB$
$$sinC + sinD = 2sin\frac{C+D}{2}.cos\frac{C-D}{2}, \quad sinC - sinD = 2cos\frac{C+D}{2}.sin\frac{C-D}{2}$$
$$cosC + cosD = 2cos\frac{C+D}{2}.cos\frac{C-D}{2}, \quad cosC - cosD = -2sin\frac{C+D}{2}.sin\frac{C-D}{2}$$

5. $2sinA.cosB = sin(A+B) + sin(A-B), \qquad 2cosA.cosB = cos(A+B) + cos(A-B)$
$2cosA.sinB = sin(A+B) - sin(A-B), \qquad 2sinA.sinB = cos(A-B) - cos(A+B)$
$$tan(A \pm B) = \frac{(tanA \pm tanB)}{(1 \mp tanA.tanB)}$$
$$sin2A = 2sinA.cosA = \frac{2tanA}{1 + tan^2A}$$
$$cos2A = cos^2A - sin^2A = 2cos^2A - 1 = 1 - 2sin^2A = \frac{1 - tan^2A}{1 + tan^2A}$$
$$sin3A = 3sinA - 4sin^3A \qquad cos3A = 4cos^3A - 3cosA$$
$$tan2A = \frac{2tanA}{1 - tan^2A} \qquad tan3A = \frac{3tanA - tan^3A}{1 - 3tan^2A}$$

6. $sinA = \sqrt{\frac{1 - cosA}{2}}, \qquad cosA = \sqrt{\frac{1 + cosA}{2}}$

7. $log_a(mn) = log_am + log_an, \qquad log_a(\frac{m}{n}) = log_am - log_an$
$log_am^n = nlog_am, \qquad log_am = \frac{logm}{loga}$

8. $Binomial Expansion:$
$$(1+x)^n = 1 +^n C_1x^1 +^n C_2x^2 + ... + x^n$$
$$= 1 + nx + \frac{n(n-1)}{2!}x^2 + ... + x^n$$

DIFFERENTIATION FORMULAE

1. $\frac{d}{dx}(x^n) = nx^{n-1}$
2. $\frac{d}{dx}(logx) = \frac{1}{x}$
3. $\frac{d}{dx}(e^x) = e^x$
4. $\frac{d}{dx}(a^x) = a^x loga$
5. $\frac{d}{dx}(sinx) = cosx$
6. $\frac{d}{dx}(cosx) = -sinx$
7. $\frac{d}{dx}(tanx) = sec^2x$
8. $\frac{d}{dx}(cotx) = -cosec^2x$
9. $\frac{d}{dx}(cosecx) = -cosecx.cotx$
10. $\frac{d}{dx}(secx) = secx.tanx$
11. $\frac{d}{dx}(sin^{-1}x) = \frac{1}{\sqrt{1-x^2}}$
12. $\frac{d}{dx}(cos^{-1}x) = -\frac{1}{\sqrt{1-x^2}}$
13. $\frac{d}{dx}(tan^{-1}x) = \frac{1}{1+x^2}$
14. $\frac{d}{dx}(cot^{-1}x) = -\frac{1}{1+x^2}$
15. $\frac{d}{dx}(sec^{-1}x) = \frac{1}{x\sqrt{x^2-1}}$
16. $\frac{d}{dx}(cosec^{-1}x) = -\frac{1}{x\sqrt{x^2-1}}$
17. $\frac{d}{dx}(uv) = u\frac{dv}{dx} + v\frac{du}{dx}$
18. $\frac{d}{dx}(\frac{u}{v}) = \frac{v\frac{du}{dx} - u\frac{dv}{dx}}{v^2}$

INTEGRATION FORMULAE

1. $\int x^n dx = \frac{x^{n+1}}{n+1}, (n \neq -1)$
2. $\int \frac{1}{x} dx = log|x|$
3. $\int e^x dx = e^x$
4. $\int a^x dx = \frac{a^x}{loga}, a > 0, a \neq 1$
5. $\int sinx dx = -cosx$
6. $\int cosx dx = sinx$
7. $\int tanx dx = -logcosx = logsecx$
8. $\int cotx dx = logsinx$
9. $\int secx dx = log(secx + tanx)$
10. $\int cosecx dx = log(cosecx - cotx)$
11. $\int sec^2x dx = tanx$
12. $\int cosec^2x dx = -cotx$

13. $\int secxtanxdx = secx$

14. $\int cosecxcotxdx = -cosecx$

15. $\int \frac{1}{\sqrt{a^2-x^2}}dx = sin^{-1}\left(\frac{x}{a}\right)$

16. $\int \frac{1}{\sqrt{x^2-a^2}}dx = log(x+\sqrt{x^2-a^2}) = cosh^{-1}(\frac{x}{a})$

17. $\int \frac{1}{\sqrt{x^2+a^2}}dx = log(x+\sqrt{x^2+a^2}) = sinh^{-1}(\frac{x}{a})$

18. $\int \frac{1}{a^2-x^2}dx = \frac{1}{2a}log\left(\frac{a+x}{a-x}\right) = \frac{1}{a}tanh^{-1}\left(\frac{x}{a}\right)$

19. $\int \frac{1}{x^2-a^2}dx = \frac{1}{2a}log\left(\frac{x-a}{x+a}\right) = -\frac{1}{a}coth^{-1}\left(\frac{x}{a}\right)$

20. $\int \frac{1}{a^2+x^2}dx = \frac{1}{a}tan^{-1}\left(\frac{x}{a}\right)$

21. $\int \sqrt{a^2-x^2}dx = \frac{x}{2}\sqrt{a^2-x^2} + \frac{a^2}{2}sin^{-1}\left(\frac{x}{a}\right)$

22. $\int \sqrt{a^2+x^2}dx$
$= \frac{x}{2}\sqrt{a^2+x^2} + \frac{a^2}{2}log(x+\sqrt{x^2+a^2})$

23. $\int \sqrt{x^2-a^2}dx$
$= \frac{x}{2}\sqrt{x^2-a^2} - \frac{a^2}{2}log(x+\sqrt{x^2-a^2})$

24. $\int e^{ax}sinbxdx = \frac{e^{ax}}{a^2+b^2}(asinbx - bcosbx)$

25. $\int e^{ax}cosbxdx = \frac{e^{ax}}{a^2+b^2}(acosbx + bsinbx)$

26. $\int uvdx = u\int vdx - \int\left(\frac{du}{dx}\int vdx\right)dx$

27. $\int [f(x)]^n f'(x)dx = \frac{[f(x)]^{n+1}}{n+1}, n \neq -1$

28. $\int \frac{f'(x)}{f(x)}dx = log|f(x)|$

29. $\int e^{f(x)}f'(x)dx = e^{f(x)}$

30. $\int e^x[f(x)+f'(x)]dx = e^x f(x)$

31. $\int sin[f(x)]f'(x)dx = -cosf(x)$

32. $\int cos[f(x)]f'(x)dx = sinf(x)$

33. $\int_0^a f(x)dx = \int_0^a f(a-x)dx$

34. $\int_0^{2a} f(x)dx = \int_0^a f(x)dx + \int_0^a f(2a-x)dx$

35.
$$\int_{-a}^{a} f(x)dx = \begin{cases} 2\int_0^a f(x)dx & ,iff(x)iseven \\ 0 & ,iff(x)isodd \end{cases}$$

36.
$$\int_0^{2a} f(x)dx = \begin{cases} 2\int_0^a f(x)dx, iff(x) = f(2a-x) \\ 0, iff(x) = f(x) = -f(2a-x) \end{cases}$$

Generalised formula for integration by parts $\int uvdx = uv_1 - u'v_2 + u''v_3 - u'''v_4 + \ldots\ldots\ldots$

where $u' = \frac{du}{dx}, u'' = \frac{d^2u}{dx^2}, u''' = \frac{d^3u}{dx^3}\ldots\ldots$ **and** $v_1 = \int vdx, v_2 = \int v_1dx, v_3 = \int v_2dx, \ldots\ldots$